Sl. No.	Author, year	n	Name of study group	Preventive measure	Primary outcome	Age (years)	Duration of anesthesia/ surgery (minutes)	BCP°	Surgery	HTN/DM/Other (n)	Type of anesthesia	ASA PS	Induction agent	Maintenance agent	Local anesthetic drug of ISB	Incidence of bradycardia, n(%)	Incidence of hypotension, n(%)	Vasoactive agent	Vasoactive agent, bolus vs infusion	Definition of bradycardia (HR, beats/min) or hypotension (mm Hg).	Definition of CDE (%).	Jadad score
1	Cho et al., 2013 [20]	15	Saline	Supportive stockings	Vasopressin prophylaxis for BCP induced hypotension	62 (39-73)	199 ± 59	65-75°	Shoulder arthroscopy	5/3/NP	GA, TIVA propofol	I-III	Propofol	Air, propofol, remifentanil	Nil	Nil	10 (66%)	Eph	Bolus	MAP < 50	rSO$_2$ > 20% or greater from BL for > 15 sec	4
		15	AVP	Supportive stockings, AVP		61 (48-70)	221 ± 52	65-75°		5/3/NP	GA, TIVA propofol	I-III	Propofol	Air, propofol, remifentanil	Nil	Nil	2 (13.3%)	Eph	Eph bolus and AVP infusion			
2	Cho et al., 2015 [21]	15	Control, saline	Supportive stockings, ringers solution	Vasopressin effect on BP	58 ± 10	168 ± 53	65-75°	shoulder arthroscopy	4/5/NP	GA, TIVA propofol	I-III	Propofol	Propofol, remifentanil	Nil	Nil	11 (73.3%)	Eph, Phe	Bolus	MAP < 50	rSO$_2$ > 20% or greater from BL for > 15 sec	2
		15	AVP, 0.025 U/kg	Supportive stockings, ringers solution, AVP		62 ± 7	147 ± 66	65-75°		5/2/unclear	GA, TIVA propofol	I-III	Propofol	Propofol, remifentanil	Nil	Nil	6 (40%)	Eph, Phe	Bolus and AVP infusion			
		15	AVP, 0.05 U/kg	Supportive stockings, ringers solution, AVP		65 ± 8	186 ± 59	65-75°		7/1/unclear	GA, TIVA propofol	I-III	Propofol	Propofol, remifentanil	Nil	Nil	4 (26.7%)	Eph, Phe	Bolus and AVP infusion			
		15	AVP, 0.075 U/kg	Supportive stockings, ringers solution, AVP		60 ± 8	187 ± 51	65-75°		5/3/unclear	GA, TIVA propofol	I-III	Propofol	Propofol, remifentanil	Nil	Nil	2 (13.3%)	Eph, Phe	Bolus and AVP infusion			
3	Frey et al., 2018 [22]	19	Ringers acetate	Nil	Volume therapy for BCP induced hemodynamic changes	52 ± 16.5	NP	70°	Shoulder surgery	NP	GA	I-II	Propofol	Propofol, sufentanil	Nil	Nil	0 (0%)	NE	Bolus and infusion	MAP < 80 or > 20% decrease from BL	NP	5
		20	HES 130/0.4	Nil		56 ± 15.6	NP	70°		NP	GA	I-II	Propofol	Propofol, sufentanil	Nil	Nil	0 (0%)	NE	Bolus and infusion			
4	Ghandour et al., 2017 [23]	110	Study	SCD	Effect of BCP angle on cerebral desaturation	NP	NP	60°	shoulder arthroscopy	NP	GA + ISB	I-II	Propofol, remifentanil	Sevoflurane, remifentanil	0.5% ropivacaine, 30ml	Nil	NP	Phe, Eph	Bolus	NP	rSO$_2$ > 20% or greater from BL	2
		110	Control	SCD		NP	NP	90°		NP	GA + ISB	I-II	Propofol, remifentanil	Sevoflurane, remifentanil	0.5% ropivacaine, 30ml	Nil	NP	Phe, Eph	Bolus			
5	Golz et al.,2020 [24]	33	Control	SCD	Effect of CS on cerebral desaturation	56.8 ± 9.4	120.3 ± 41	60-65°	shoulder arthroscopy	20/10/5	GA + ISB/SCB/ ICB	NP	Propofol	Sevoflurane (Desflurane, N$_2$O)	0.5% bupivacaine or 1.3% liposomal bupivacaine	NP	NP	Phe, Eph, fluid bolus	Bolus	NP	rSO$_2$ > 20% or greater from BL	3
		33	CS	SCD		49.7 ± 12.6	117.4 ± 37.2	60-65°		14/8/13	GA + ISB/SCB/ ICB	NP	Propofol	Sevoflurane (Desflurane, N$_2$O)	0.5% bupivacaine or 1.3% liposomal bupivacaine	NP	NP	Phe, Eph, fluid bolus	Bolus			
6	Jang et al., 2017 [25]	15	PR-Saline	Nil	Effects of anesthetics and AVP on hemodynamics and cerebral oxygenation	62 (46-70)	151 ± 45	65-75°	Shoulder surgery	5/4/NP	GA, TIVA Propofol	I-III	Propofol, remifentanil	Propofol, remifentanil	Nil	Nil	11 (73%)	Eph	Bolus	MAP < 50	rSO2 > 20% or greater from BL for > 15 sec	2
		15	PR-AVP	AVP, 0.05 U/kg bolus		59 (53-78)	161 ± 29	65-75°		7/1/NP	GA, TIVA Propofol	I-III	Propofol, remifentanil	Propofol, remifentanil	Nil	Nil	4/16 (25%)	Eph	Bolus and AVP loading dose			
		15	SN-Saline	Nil		62 (45-77)	181 ± 56	65-75°		4/2/NP	GA	I-III	Propofol, remifentanil	N$_2$O, sevoflurane	Nil	Nil	6 (40%)	Eph	Bolus			
		15	SN-AVP	AVP, 0.05 U/kg bolus		65 (51-73)	173 ± 34	65-75°		5/2/NP	GA	I-III	Propofol, remifentanil	N$_2$O, sevoflurane	Nil	Nil	3 (20%)	Eph	Bolus			
7	Kwak et al., 2011 [26]	33	Control	6% HES preload	Effect of SCD on cerebral oxygenation	46 ± 12	NP	70°	Shoulder arthroscopy	0/0/NP	GA	I-II	Propofol, remifentanil	Sevoflurane, remifentanil	Nil	Nil	8 (24.2%)	Eph	Bolus	MBP < 60 or > 20% decrease from BL	NP	4
		33	SCD	6% HES preload		42 ± 12	NP	70°		0/0/NP	GA	I-II	Propofol, remifentanil	Sevoflurane, remifentanil	Nil	Nil	1 (3.03%)	Eph	Bolus			
8	Kwak et al., 2017 [27]	20	Normocapnia	6% HES preload	Hypercapnia effect on lung oxygenation	53 ± 8	NP	70°	Shoulder arthroscopy	NP	GA	I-II	Propofol, alfentanil	Sevoflurane	Nil	Nil	9 (45%)	Phe, Eph	Bolus	MBP < 60	rSO$_2$ ≥ 20% or greater from BL for ≥ 15 sec	2
		20	Hypercapnia	6% HES preload		50 ± 9	NP	70°		NP	GA	I-II	Propofol, alfentanil	Sevoflurane	Nil	Nil	7 (35%)	Phe, Eph	Bolus			
9	Murphy et al., 2014 [28]	34	Study	Nil	Effect of ventilation on cerebral oxygenation	54.7 ± 12.4	98 (75-185)	80-90°	Shoulder arthroscopy	12/5/9	GA	I-III	Propofol, fentanyl	Sevoflurane, fentanyl	Nil	Nil	34 (100%)	Phe	Bolus	MBP >20% decrease from BL	rSO$_2$ ≥ 20% or greater from BL	4
		36	Control	Nil		52.3 ± 13.8	96 (40-193)	80-90°		15/5/8	GA	I-III	Propofol, fentanyl	Sevoflurane, fentanyl	Nil	Nil	36 (100%)	Phe	Bolus			
10	Oh CS et al., 2019 [29]	34	Control	Crystalloid hydration (bodyweight*5ml)	Effect of regional ischemic preconditioning on cerebral oxygenation	60 (12)	166 (48)	NP	Shoulder surgery	11/6/NP	GA	NP	Thiopentone sodium	Sevoflurane, remifentanil	Nil	9/34	NP	Phe, Eph	Bolus	MBP >20% decrease from BL	rSO$_2$ ≥ 20% or greater from preanesthetic value or absolute value of < 40% for > 1 minute	5
		29	Study	Crystalloid hydration (bodyweight*5ml)		60 (16)	177 (48)	NP		8/1/NP	GA	NP	Thiopentone sodium	Sevoflurane, remifentanil	Nil	5/29	NP	Phe, Eph	Bolus			
11	Soeding et al., 2013 [7]	17	Saline	Prehydration with Hartmann solution	Effect of phenylephrine on cerebral oxygen saturation	40 (32-48)	NP	>75°	Shoulder surgery	NP	GA + ISB	I-II	Propofol, fentanyl	Sevoflurane	0.75% ropivacaine, 30-35ml	Nil	4 (25%)	Metaraminol	Bolus	MBP < 65	rSO$_2$ ≥ 25% or greater from BL or absolute value < 50%	2
		17	Phe	Prehydration with Hartmann solution, Phe infusion		47 (37-57)	NP	>75°		NP	GA + ISB	I-II	Propofol, fentanyl	Sevoflurane	0.75% ropivacaine, 30-35ml	Unclear	2 (11%)	Metaraminol	Bolus			
12	Woo et al., 2018 [13]	19	Control	NP	Effect of compression stockings on CDEs	47 ± 20	164.4 ± 38.5	45°	Shoulder surgery	3/2/unclear	GA + ISB	I-III	Propofol, fentanyl	Sevoflurane, Air-oxygen	0.5% ropivacaine, 12ml	Nil	12 (63.2%)	Eph	Bolus	MBP < 65 or > 20% decrease from BL	rSO$_2$ ≥ 20% or greater from BL for ≥ 15 sec	3
		19	CS group	NP		48 ± 15	167.1 ± 29.5	45°		5/2/unclear	GA + ISB	I-III	Propofol, fentanyl	Sevoflurane, Air-oxygen	0.5% ropivacaine, 12ml	Nil	4 (21.1%)	Eph	Bolus			